

Lab

Accessing databases

Lab: Java Program with MySQL CRUD Operations

Objective

In this lab, you will build a simple Java application that connects to a MySQL database and performs basic **CRUD** operations — **Create**, **Read**, **Update**, and **Delete**. By the end, you will have a working program that interacts with a real database and reinforces your understanding of JDBC and MySQL.

Setup and implementation

- Open MySQL Workbench.
- Open a New SQL TAB
- Paste the text from the QASTore.sql (see the resources folder) in a new query and then execute the query.

Step 3: Set up your Java project

- Open **Eclipse** and create a new **Maven project**.
- In your pom.xml, add the MySQL Connector dependency. You can find it on mvnrepository.com.

```
<dependency>
```

```
    <groupId>mysql</groupId>
```

```
    <artifactId>mysql-connector-java</artifactId>
```

```
    <version>8.0.33</version> <!-- Use the latest stable version -->
```

```
</dependency>
```

Step 4: Write your Java program

- Use JDBC to connect to your local MySQL server.
- Implement the following operations:
 - **Create**: Insert a new record into your table.
 - **Read**: Retrieve and display records.
 - **Update**: Modify an existing record.

- **Delete:** Remove a record.
- Use the code on the slides to start with.

Stretch goal: Use PreparedStatement

- Refactor your code to use PreparedStatement instead of string concatenation.
- This improves security (prevents SQL injection) and makes your code more robust and maintainable.



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